

SQL Code

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1  drop database joinsdb ;
2  create database joinsdb ;
3  use joinsdb ;
4
5  CREATE TABLE departments (
6  dept_id INT PRIMARY KEY,
7  dept_name VARCHAR(50),
8  location VARCHAR(50)
9  );
10 INSERT INTO departments VALUES
11 (10,'HR','Jaipur'),
12 (20,'IT','Bangalore'),
13 (30,'Finance','Mumbai'),
14 (40,'Marketing','Delhi'),
15 (50,'Legal','Pune');
16 CREATE TABLE employees (
17 emp_id INT PRIMARY KEY,
18 emp_name VARCHAR(50),
19 dept_id INT,
20 salary INT,
21 city VARCHAR(50)
22 );
23 INSERT INTO employees VALUES
24 (101,'Amit',10,45000,'Jaipur'),
25 (102,'Bhavna',20,72000,'Bangalore'),
26 (103,'Chirag',20,68000,'Bangalore'),
27 (104,'Divya',30,55000,'Mumbai'),
28 (105,'Esha',NULL,39000,'Delhi'),
29 (106,'Farhan',40,61000,'Delhi'),
30 (107,'Gaurav',60,50000,'Kolkata'),
31 (108,'Hina',30,83000,'Mumbai');
32 CREATE TABLE projects (
33 proj_id INT PRIMARY KEY,
34 proj_name VARCHAR(50),
35 dept_id INT,
36 budget INT
37 );
38 INSERT INTO projects VALUES
39 (1,'Payroll Revamp',10,120000),
40 (2,'Cloud Migration',20,500000),
41 (3,'Mobile App',20,300000),
42 (4,'Audit Automation',30,250000),
43 (5,'Brand Refresh',40,180000),
44 (6,'Data Lake',70,400000);
45
46
47 select * from depart ;
48 select * from empl;
49 select * from proje bcts ;
50 use joinsdb ;
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51
52 -- Q1. List each employee with their department name and location.
53 select e.emp_id , e.emp_name , d.dept_name , d.location from empl as e
54 inner join depart as d
55 on e.dept_id = d.dept_id ;
56
57
58 -- Q2. Show every project along with the name of the department that owns it.
59 select p.proj_id , p.proj_name , d.dept_name , p.budget from projects as p
60 inner join depart as d
61 on p.dept_id = d.dept_id ;
62
63
64 -- Q3. Show employee names together with the projects run by their own department.
65 select e.emp_name , dept_name , p.proj_name from empl as e
66 inner join depart as d
67 on e.dept_id = d.dept_id
68 inner join projects as p
69 on d.dept_id = p.dept_id ;
70
71
72 -- Q4. List ALL employees with their department name; show NULL when the employee has
73 no matching department.
74 select e.emp_id , e.emp_name , e.dept_id , d.dept_name from empl as e
75 left join depart as d
76 on e.dept_id = d.dept_id ;
77
78 -- Q5. list all departments with their projects; departments with no project must
79 still appear.
80 select d.dept_id, d.dept_name, p.proj_name, p.budget
81 from depart as d
82 left join projects as p
83 on d.dept_id = p.dept_id;
84
85 -- Q6. find only those employees who do not belong to any valid department.
86 select e.emp_id, e.emp_name, e.dept_id
87 from empl as e
88 left join depart as d
89 on e.dept_id = d.dept_id
90 where d.dept_id is null;
91
92
93 -- Q7. list all departments and any employees in them, using a right join with
94 employees on the left.
95 select e.emp_name, d.dept_id, d.dept_name
96 from empl as e
97 right join depart as d
98 on e.dept_id = d.dept_id;
99
100 -- Q8. list all projects and the department that owns them, keeping projects that
101 point to a missing department.
102 select d.dept_name, p.proj_id, p.proj_name, p.dept_id

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101     from depart as d
102     right join projects as p
103     on d.dept_id = p.dept_id;
104
105
106     -- Q9. find departments that currently have no employee assigned.
107     select d.dept_id, d.dept_name, d.location
108     from depart as d
109     left join empl as e
110     on d.dept_id = e.dept_id
111     where e.emp_id is null;
112
113     -- Q10. produce every possible pairing of the finance/legal departments with employees
114     earning above 70000.
115     select e.emp_name, e.salary, d.dept_name
116     from empl as e
117     cross join depart as d
118     where e.salary > 70000
119     and d.dept_name in ('Finance','Legal');
120
121     -- Q11. pair every project having a budget of at least 400000 with every department
122     located in mumbai or pune.
123     select p.proj_name, p.budget, d.dept_name, d.location
124     from projects as p
125     cross join departments as d
126     where p.budget >= 400000
127     and d.location in ('Mumbai','Pune');
128
129     -- Q12. how many total row combinations result from cross join of employees and
130     departments?
131     select count(*) as total_combinations
132     from empl as e
133     cross join depart as d;
134
135     -- Q13. show employees whose salary is greater than 60000.
136     select emp_id, emp_name, dept_id, salary, city
137     from empl
138     where salary > 60000;
139
140     -- Q14. show all employees based in delhi or mumbai.
141     select emp_id, emp_name, salary, city
142     from empl
143     where city in ('Delhi','Mumbai');
144
145     -- Q15. show employees whose dept_id is null (unassigned employees).
146     select emp_id, emp_name, dept_id, salary
147     from empl
148     where dept_id is null;
149
150     -- Q16. show departments located in bangalore or delhi.
151     select dept_id, dept_name, location
152     from depart
153     where location in ('Bangalore','Delhi');

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152 -- Q17. show departments whose dept_id is 30 or higher.
153 select dept_id, dept_name, location
154 from depart
155 where dept_id >= 30;
156
157 -- Q18. show departments whose name starts with the letter 'f' or 'l'.
158 select dept_id, dept_name, location
159 from depart
160 where dept_name like 'F%'
161 or dept_name like 'L%';
162
163 -- Q19. show projects with a budget between 200000 and 400000 (inclusive).
164 select proj_id, proj_name, dept_id, budget
165 from projects
166 where budget between 200000 and 400000;
167
168 -- Q20. show all projects that belong to dept_id 20.
169 select proj_id, proj_name, dept_id, budget
170 from projects
171 where dept_id = 20;
172
173 -- Q21. show projects whose name contains the word 'a' and budget is under 300000.
174 select proj_id, proj_name, budget
175 from projects
176 where proj_name like '%a%'
177 and budget < 300000;
178
179
180 -- Q22. show employee name, department name, location and project name for all
    matching rows across all three tables.
181 select e.emp_name, d.dept_name, d.location, p.proj_name, p.budget
182 from empl as e
183 inner join depart as d
184 on e.dept_id = d.dept_id
185 inner join projects as p
186 on d.dept_id = p.dept_id;
187
188
189 -- Q23. show all employees, plus department and project details where available (keep
    employees even with no dept/project).
190 select e.emp_id, e.emp_name, d.dept_name, p.proj_name
191 from empl as e
192 left join depart as d
193 on e.dept_id = d.dept_id
194 left join projects as p
195 on d.dept_id = p.dept_id;
196
197
198 -- Q24. show employees earning more than 60000 along with department and any project
    over 250000 budget.
199 select e.emp_name, e.salary, d.dept_name, p.proj_name, p.budget
200 from empl as e
201 inner join depart as d
202 on e.dept_id = d.dept_id

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203     inner join projects as p
204     on d.dept_id = p.dept_id
205     where e.salary > 60000
206     and p.budget > 250000;
207
208
209 -- Q25. list all departments with their employees and projects, including departments
    having neither.
210 select d.dept_id, d.dept_name, e.emp_name, p.proj_name
211 from depart as d
212 left join empl as e
213 on d.dept_id = e.dept_id
214 left join projects as p
215 on d.dept_id = p.dept_id;
216
217
218 -- Q26. show employees who work in a department located in bangalore or mumbai, along
    with the projects of that department.
219 select e.emp_name, d.location, p.proj_name
220 from empl as e
221 inner join depart as d
222 on e.dept_id = d.dept_id
223 inner join projects as p
224 on d.dept_id = p.dept_id
225 where d.location in ('Bangalore', 'Mumbai');
226
227
228 -- Q27. pair employees who work in the same city (avoid duplicate pairs and self-
    pairing).
229 select e1.emp_name as employee_1,
230 e2.emp_name as employee_2,
231 e1.city
232 from empl as e1
233 inner join empl as e2
234 on e1.city = e2.city
235 and e1.emp_id < e2.emp_id;
236
237
238 -- Q28. show every project along with employees of that department; keep projects with
    no employees.
239 select p.proj_name, p.dept_id, e.emp_name
240 from projects as p
241 left join empl as e
242 on p.dept_id = e.dept_id;
243
244
245
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